analyzing its implications discovered an unexpected link between general relativity and electromagnetism. Thus, although it was a great discovery in its own right, it lacked a sense of inevitability. If you asked Kaluza and Klein *why* the universe had five spacetime dimensions rather than four, or six, or seven, or 7,000 for that matter, they wouldn't have had an answer much more convincing than "Why not?"

More than three decades later, the situation changed radically. String theory is the first approach to merge general relativity and quantum mechanics; moreover, it has the potential to unify our understanding of all forces and all matter. But the quantum mechanical equations of string theory don't work in four spacetime dimensions, nor in five, six, seven, or 7,000. Instead, for reasons discussed in the next section, the equations of string theory work only in ten spacetime dimensions—nine of space, plus time. String theory *demands* more dimensions.

This is a fundamentally different kind of result, one never before encountered in the history of physics. Prior to strings, no theory said anything at all about the number of spatial dimensions in the universe. Every theory from Newton to Maxwell to Einstein assumed that the universe had three space dimensions, much as we all assume the sun will rise tomorrow. Kaluza and Klein proffered a challenge by suggesting that there were four space dimensions, but this amounted to yet another assumption—a different assumption, but an assumption nonetheless. Now, for the first time, string theory provided equations that *predicted* the number of space dimensions. A calculation—not an assumption, not a hypothesis, not an inspired guess—determines the number of space dimensions according to string theory, and the surprising thing is that the calculated number is not three, but nine. String theory leads us, *inevitably,* to a universe with six extra space dimensions and hence provides a compelling, ready-made context for invoking the ideas of Kaluza and Klein.

The original proposal of Kaluza and Klein assumed only one hidden dimension, but it's easily generalized to two, three, or even the six extra dimensions required by string theory. For example, in Figure 12.8a we replace the additional circular dimension of Figure 12.7, a one-dimensional shape, with the surface of a sphere, a two-dimensional shape (recall from the discussion in Chapter 8 that the surface of a sphere is